Smart Parking System for Smart Cities

Prepared For

Smart–Internz

Interest of

Things (IoT)

By

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Project Title
Smart Parking System for Smart Cities

Category
Internet of Things (IoT) | Smart City Solution | Web Application

Skills Utilized

1. HTML5
2. CSS3
3. JavaScript
4. Firebase Realtime Database
5. Git & GitHub
6. NodeMCU ESP8266
7. HC-SR04 Ultrasonic Sensor

Project Overview

Smart Parking System for Smart Cities is an Internet of Things (IoT) based parking management solution designed to improve parking efficiency in urban environments. The rapid growth of vehicles in metropolitan areas has increased the demand for intelligent parking systems that can reduce traffic congestion and minimize the time spent searching for available parking spaces.

The proposed system provides a user-friendly web interface that allows users to monitor parking slot availability in real time, reserve parking spaces, and generate QR-based parking tickets. The system utilizes Firebase Realtime Database to synchronize parking information and ensure that slot status is updated instantly across the application.

The project integrates modern web technologies including HTML, CSS, JavaScript, and Firebase to create an interactive dashboard capable of displaying parking status, vehicle statistics, entry and exit gate information, and booking details. The solution demonstrates how IoT-enabled smart parking systems can contribute towards the development of smart city infrastructure and improved urban mobility.

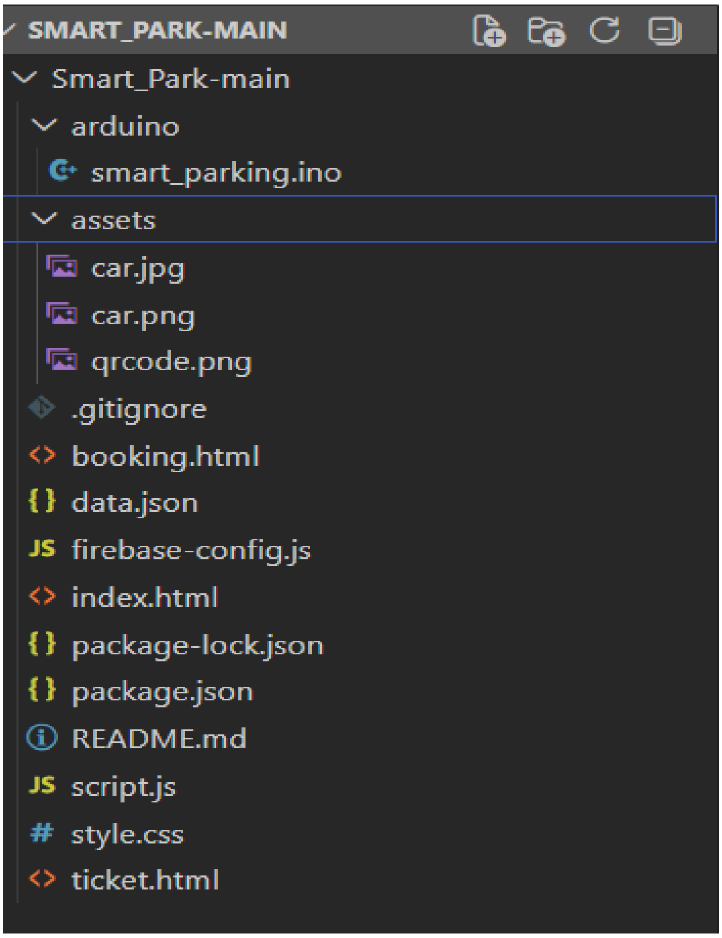
1. **Key Features**

Expand to 8–10 bullet points.

- Real-time parking slot monitoring
- Parking slot reservation system
- QR-based ticket generation
- Vehicle entry tracking
- Vehicle exit tracking
- Parking fee calculation

- Dashboard analytics
- Entry gate monitoring
- Exit gate monitoring
- Responsive user interface
- Firebase cloud integration

1. Architecture:



2. Final Project Flow

2.1 Requirement Analysis:

• Studied the problem of parking congestion in urban areas.
• Identified the need for a real-time parking management solution.
• Defined project objectives including parking slot monitoring, online booking, and ticket generation.

2.2 User Interface Development:

• Designed the Smart Parking dashboard using HTML, CSS, and JavaScript.
• Developed responsive pages for parking monitoring, booking, and ticket generation.
• Created a modern interface for improved user experience.

2.3 Firebase Realtime Database Integration:

• Configured Firebase Realtime Database.
• Established real-time communication between the dashboard and database.
• Stored parking slot information and booking records.

2.4 Parking Slot Monitoring Module:

• Implemented parking slot status management.
• Displayed slot availability as FREE, BOOKED, or OCCUPIED.
• Updated parking information dynamically on the dashboard.

2.5 Vehicle Booking Module:

• Developed a booking form for vehicle owners.
• Collected user details, vehicle number, start time, and end time.
• Validated booking information before confirmation.

2.6 Parking Fee Calculation:

• Implemented automatic parking fee calculation.
• Calculated charges based on booking duration.
• Displayed total parking charges before confirmation.

2.7 QR Ticket Generation:

• Generated a unique booking ID for every reservation.
• Created QR-based parking tickets containing booking details.
• Displayed ticket information including vehicle number, slot number, booking time, and payment details.

2.8 Vehicle Entry and Exit Monitoring:

• Implemented vehicle count monitoring.
• Displayed total vehicles entered and exited.
• Displayed entry gate and exit gate status on the dashboard.

2.9 GitHub Deployment:

• Created a GitHub repository for project management.
• Uploaded project files and documentation.
• Maintained version control throughout development.

2.10 Testing and Validation:

• Tested parking slot availability updates.
• Verified booking functionality.
• Validated QR ticket generation.
• Tested dashboard responsiveness on different screen sizes.
• Confirmed successful Firebase synchronization and data updates.

3. User Scenarios

Scenario 1: Parking Slot Reservation

User: Vehicle Owner

Objective:
To reserve an available parking slot before arriving at the parking facility.

Process:
• The user accesses the Smart Parking Dashboard.
• The system displays the current parking slot status.
• The user selects an available parking slot and proceeds to the booking page.
• The user enters personal details, vehicle number, start time, and end time.
• The system calculates the parking fee automatically based on the selected duration.
• After successful validation, the booking is confirmed and stored in Firebase Realtime Database.
• A unique booking ID and QR-based parking ticket are generated.

Outcome:
The user successfully reserves a parking slot and receives a digital parking ticket for future verification.

Scenario 2: Parking Administrator Monitoring Parking Status

User: Parking Administrator

Objective:
To monitor parking occupancy and vehicle movement in real time.

Process:
• The administrator accesses the Smart Parking Dashboard.
• The dashboard retrieves live parking data from Firebase Realtime Database.
• The administrator monitors parking slot status such as FREE, BOOKED and OCCUPIED.
• Vehicle entry and exit statistics are displayed on the dashboard.
• Entry gate and exit gate status are continuously monitored.
• Any changes in parking slot occupancy are reflected instantly on the dashboard.

Outcome:
The administrator obtains a real-time overview of parking operations and can efficiently manage parking resources.

Scenario 3: QR Ticket Verification and Parking Access

User: Registered Parking User

Objective:
To verify a parking reservation and gain access to the allocated parking slot.

Process:
• After completing the booking process, the user receives a QR-based parking ticket.
• The ticket contains booking details including booking ID, vehicle number, parking duration, and parking fee.
• The user presents the QR ticket at the parking facility.
• The booking details are verified using the ticket information.
• The system confirms the reservation and grants access to the allocated parking slot.

Outcome:
The reservation is successfully verified, and the user gains access to the reserved parking space.

4. Challenges Faced

During the development of the Smart Parking System for Smart Cities, several technical and implementation challenges were encountered.

• Understanding and implementing Firebase Realtime Database for storing and synchronizing parking information in real time.

• Designing a responsive and visually appealing dashboard capable of displaying parking status, vehicle information, and parking statistics effectively.

• Developing a reliable booking module that prevents conflicting reservations and ensures accurate parking slot allocation.

• Implementing automatic parking fee calculation based on booking duration while maintaining data accuracy.

• Integrating QR-based ticket generation and ensuring that ticket information remains consistent throughout the booking process.

• Managing parking slot state transitions between FREE, BOOKED, and OCCUPIED conditions.

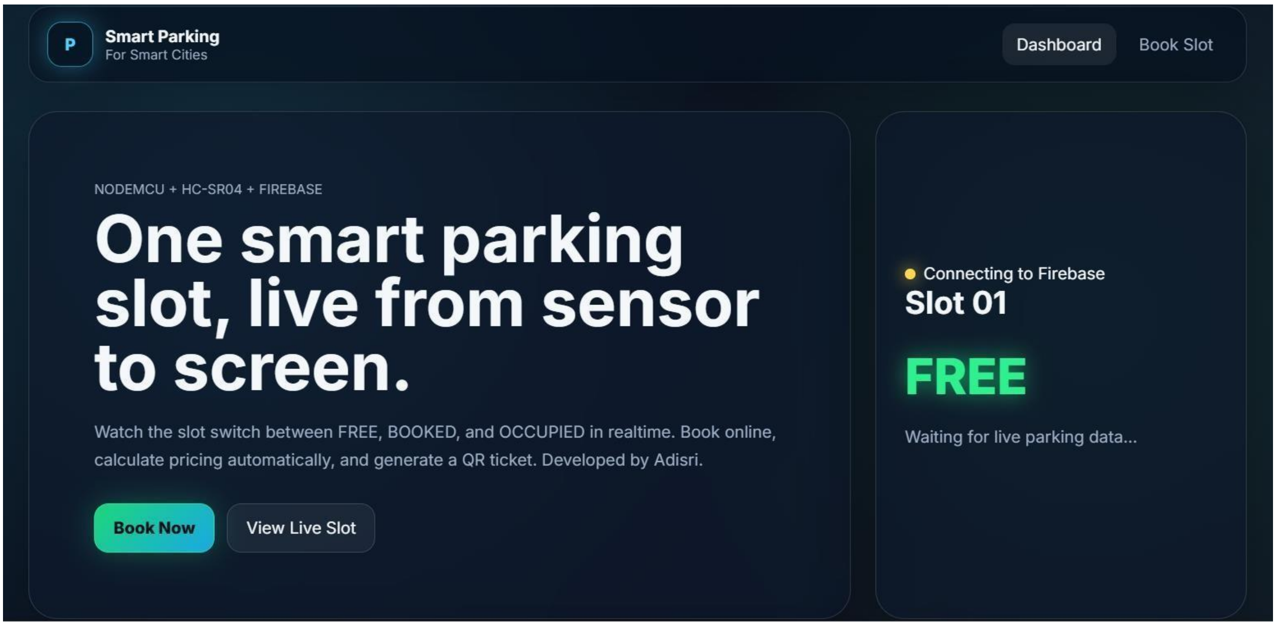
• Testing multiple booking scenarios and validating real-time updates across different pages of the application.

• Deploying the project to GitHub and maintaining proper project documentation and version control.

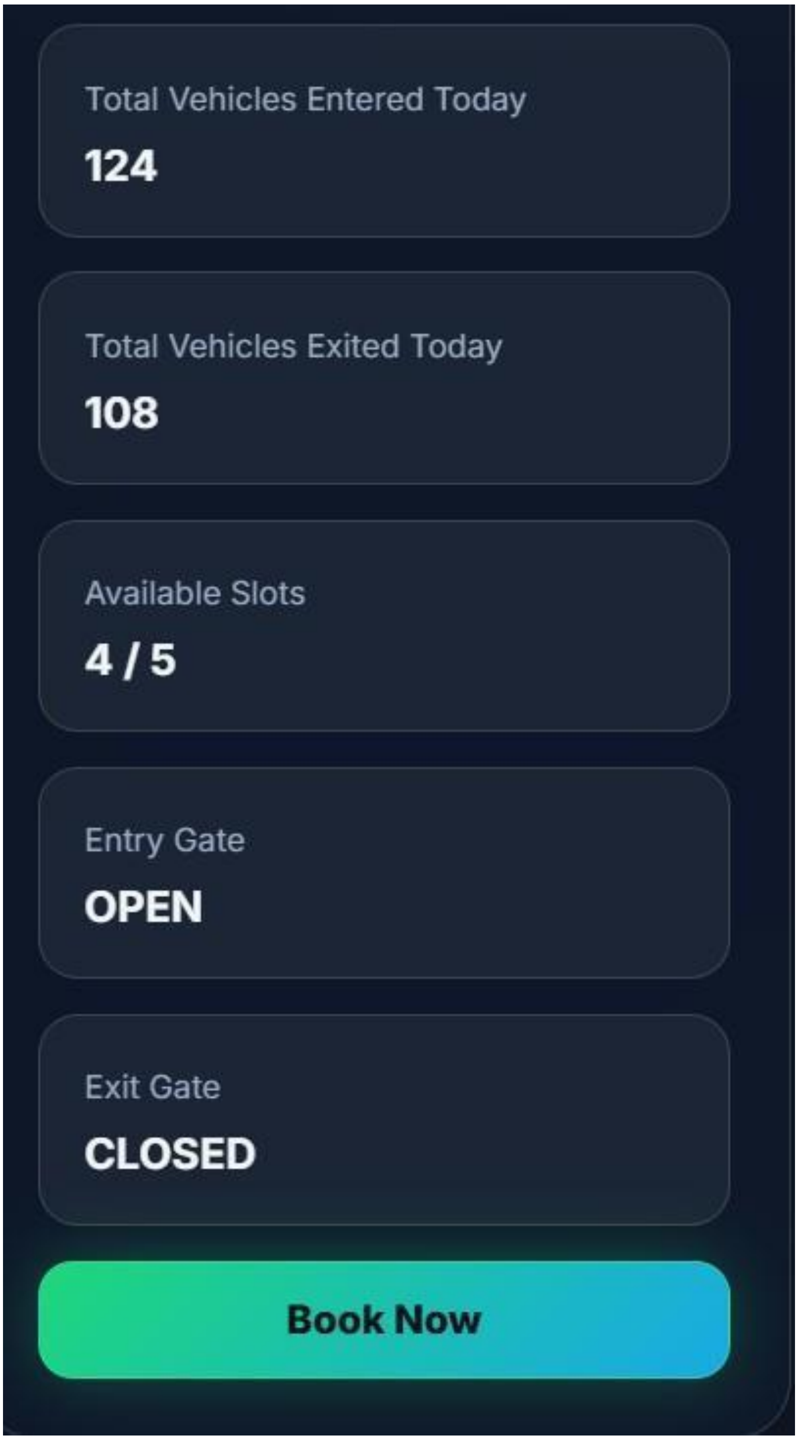
Despite these challenges, the project was successfully completed and all major functionalities were implemented and tested.

5. Output Pages

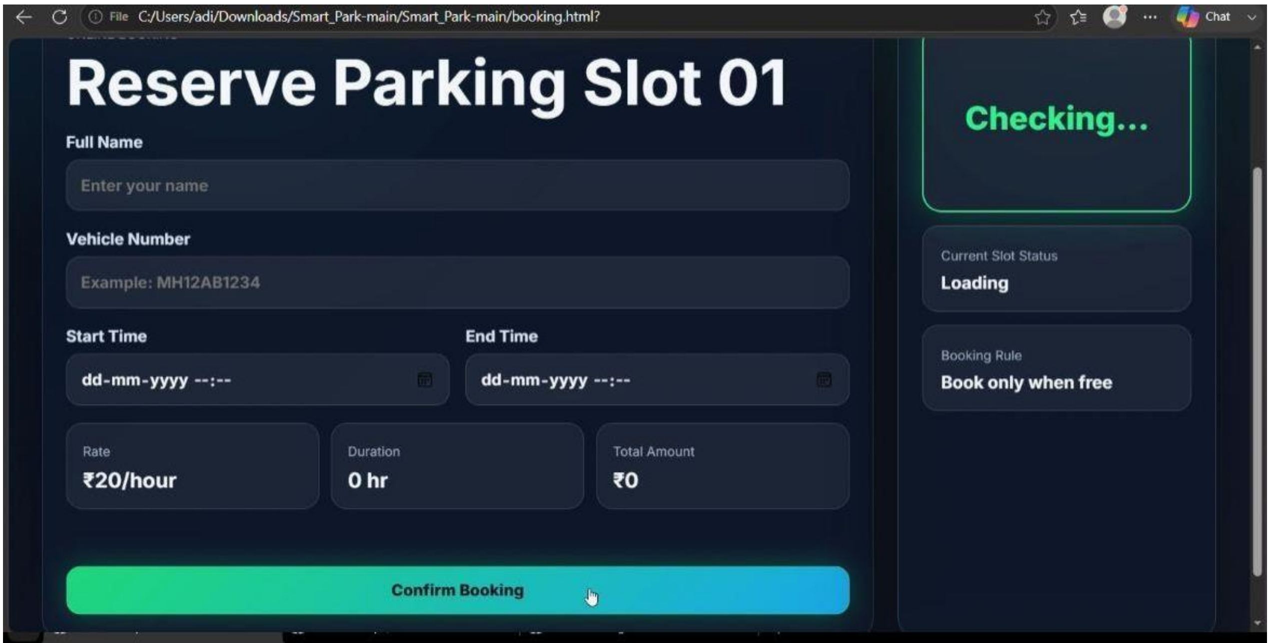
5.1 Home Dashboard

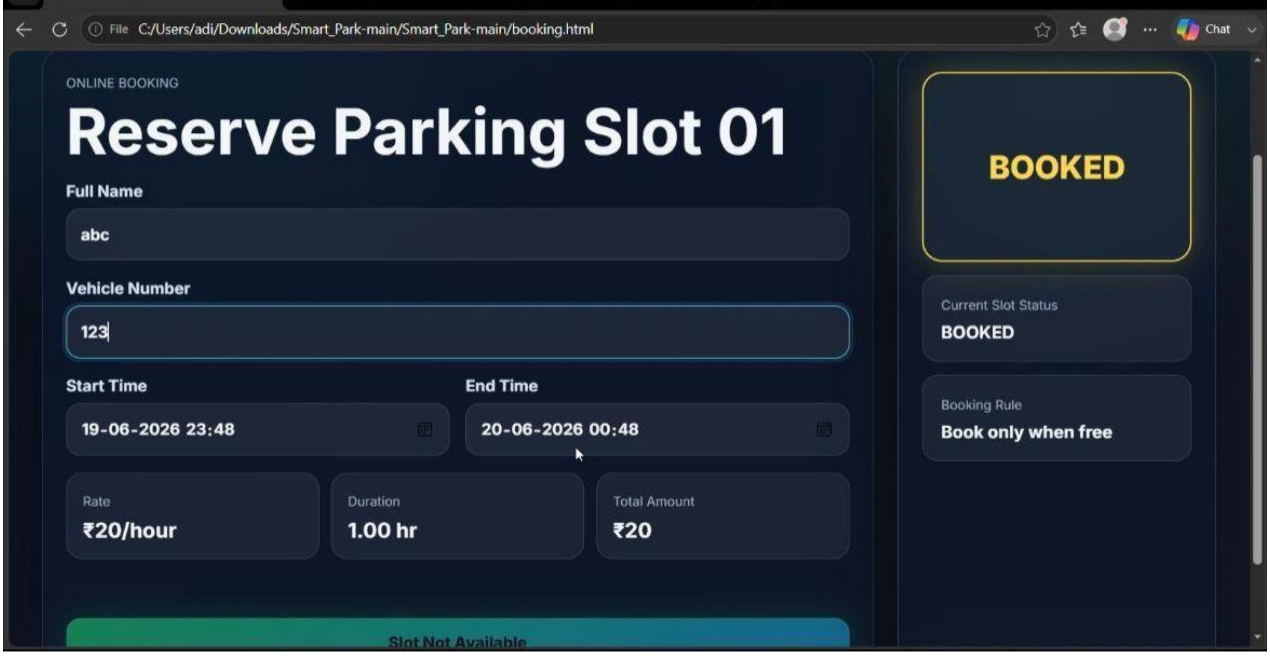


5.2 Smart Parking Analytics Panel

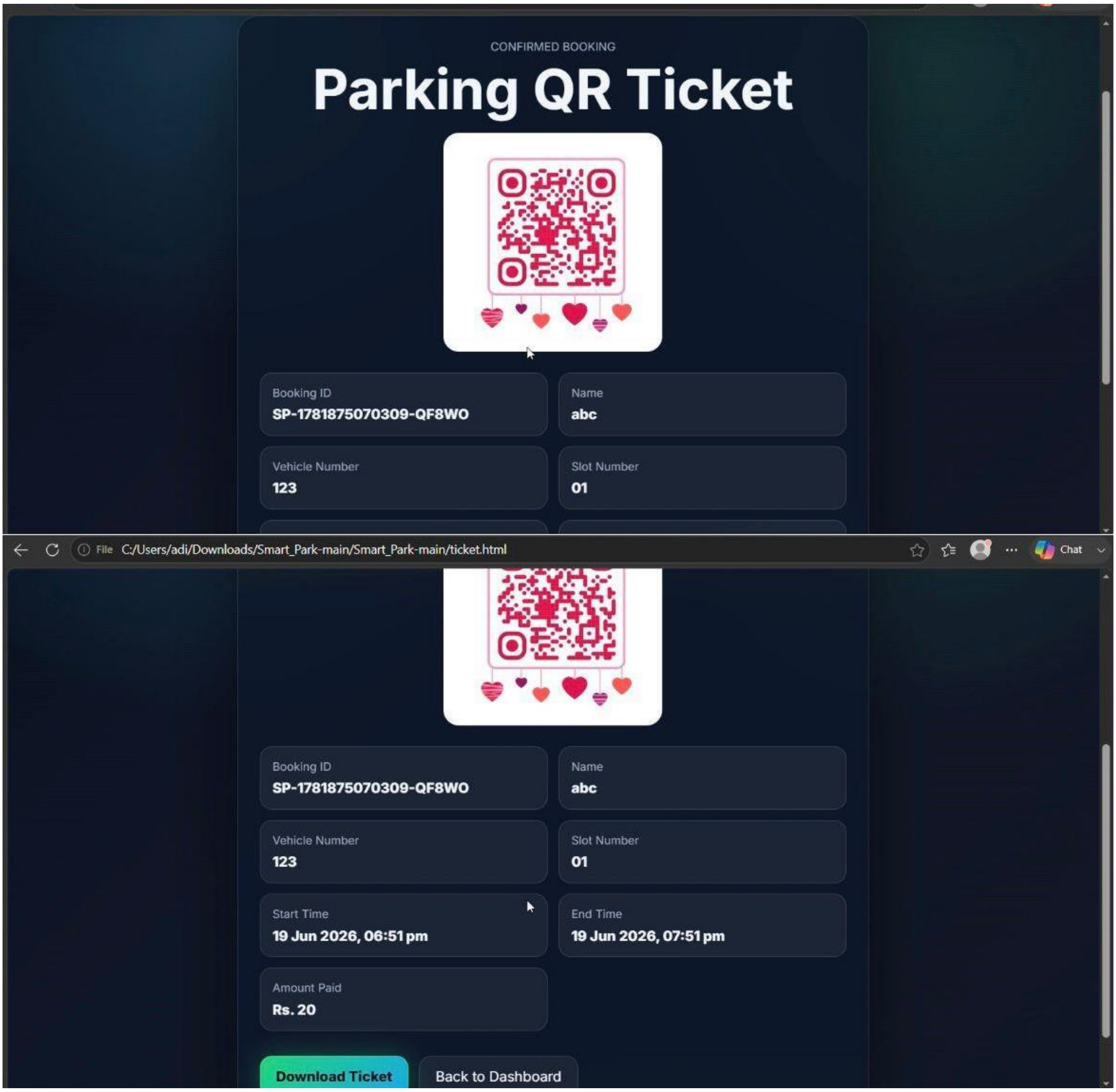


5.3 Parking Slot Reservation Space





5.4 Booking Confirmation QR Code



6. Conclusion

The Smart Parking System for Smart Cities successfully demonstrates the application of Internet of Things (IoT) concepts, cloud-based databases, and modern web technologies in solving real-world urban parking challenges. The system provides an efficient platform for monitoring parking slot availability, managing parking reservations, generating QR-based tickets, and tracking vehicle movement through an interactive dashboard.

By integrating HTML, CSS, JavaScript, and Firebase Realtime Database, the project achieves real-time synchronization of parking information while maintaining a user-friendly experience. Features such as online booking, parking fee calculation, vehicle statistics, and gate status monitoring enhance the overall functionality of the system and contribute to efficient parking management.

The project highlights how smart parking solutions can reduce traffic congestion, minimize the time spent searching for parking spaces, and improve resource utilization in urban environments. It also demonstrates the potential of IoT-enabled systems in supporting the development of smart cities and intelligent transportation infrastructure.

Future enhancements may include support for multiple parking zones, automatic vehicle detection using sensors, license plate recognition, mobile application integration, and secure online payment facilities. Overall, the project serves as a practical implementation of smart city technology and provides a scalable foundation for advanced parking management solutions.

- Git Hub Link: https://github.com/saniyanikam1205/Smart-parking-system-for-Smart-cities.git
- Demo Link: https://drive.google.com/file/d/10_o5cqt0sAfYEhkp7iNNvpNwf4nN3Pi_/view?usp=drivesdk